

Customer Shopping Behavior Analysis

An end-to-end data project examining 3,900 transactions across product categories to uncover insights on spending, customer segments, product preferences, and subscription behavior — driving strategic business decisions.



DATASET OVERVIEW

Dataset Summary

3900

Transactions

Total purchase records analyzed

18

Columns

Features per transaction

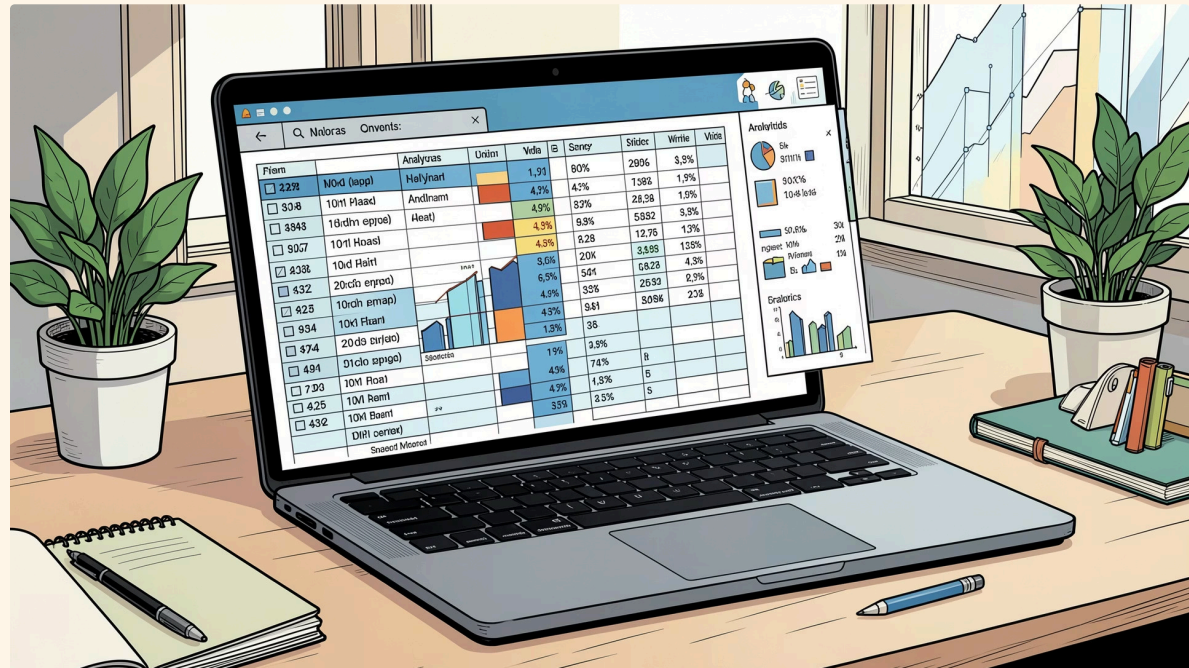
37

Missing Values

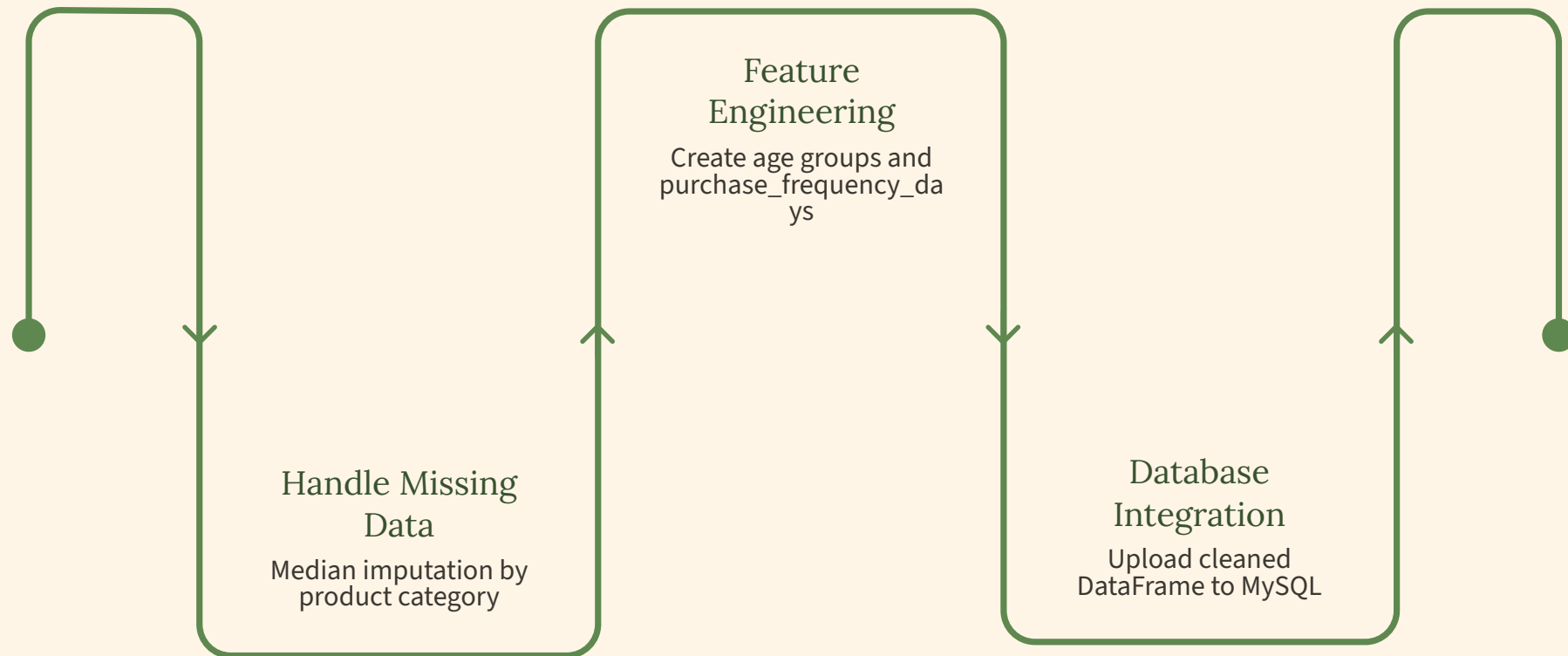
In review rating column only

Key Features

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discount, Promo Code, Frequency, Review Rating, Shipping Type

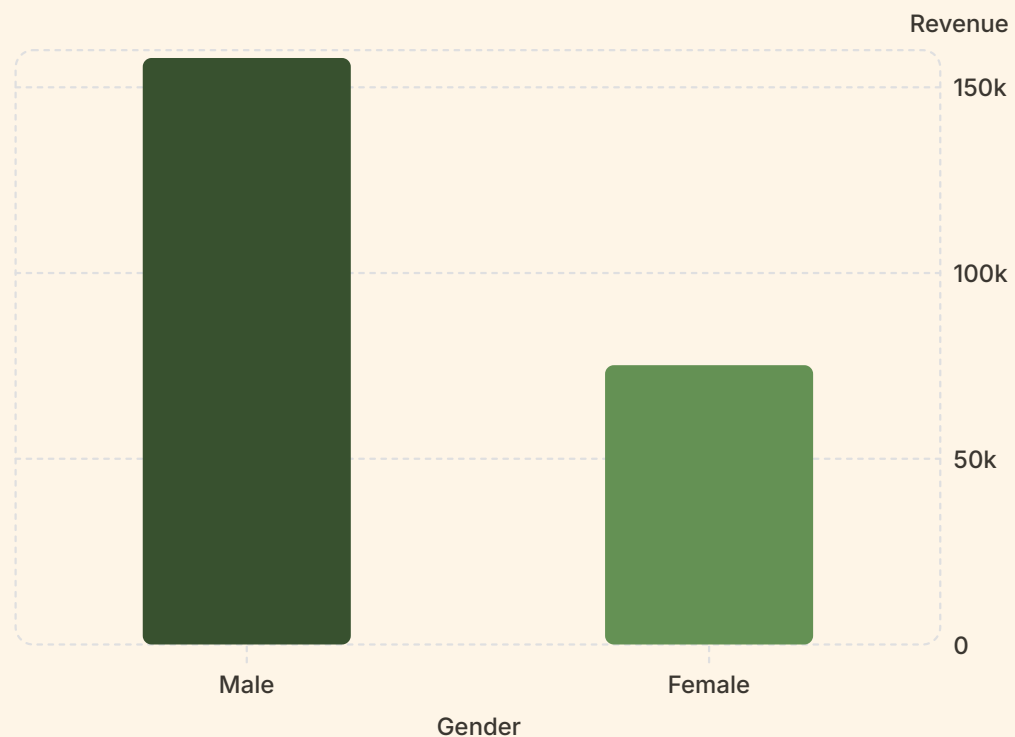


Exploratory Data Analysis in Python



A structured three-step workflow prepared the dataset: missing review ratings were imputed using the median rating per product category, new features including age groups and purchase frequency were engineered, and the cleaned DataFrame was uploaded to MySQL for SQL-based analysis.

Revenue: Male vs. Female Customers



Key Finding

Male customers generated **more than double** the revenue of female customers — \$157,890 vs. \$75,191. This significant gap suggests a key segment to target for growth.



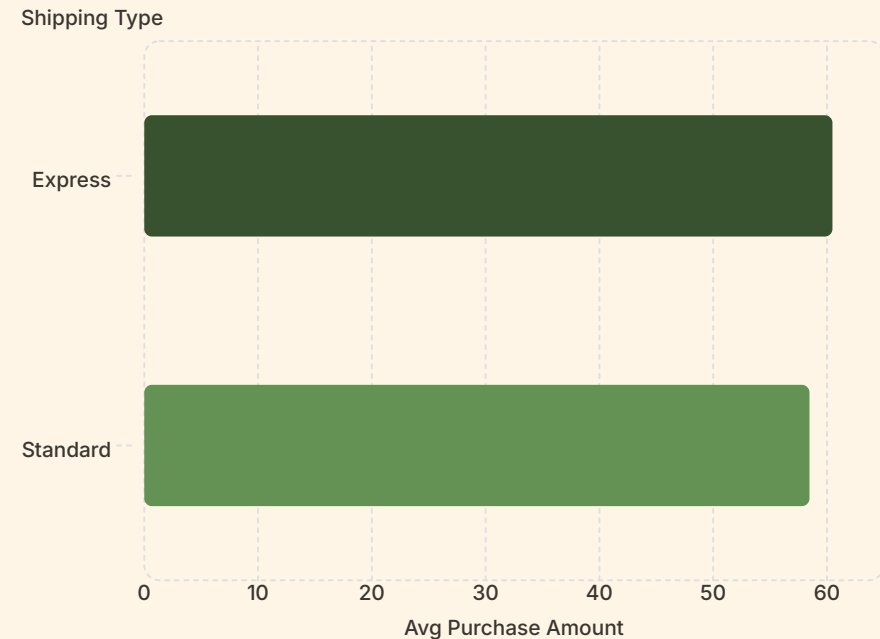
Male revenue is **2.1× higher** than female revenue across 3,900 transactions.

Discount Users & Shipping Preferences

Customers Using Discounts

Several customers who applied discounts spent **above the average** purchase amount:

Customer ID	Purchase Amount
2	64
3	73
4	90
7	85
9	97
12	68



Express shipping customers spend slightly more on average (\$60.48 vs. \$58.46).

Top 5 Best-Reviewed Products



Gloves

Avg Rating: **3.86**



Sandals

Avg Rating: **3.84**



Boots

Avg Rating: **3.82**

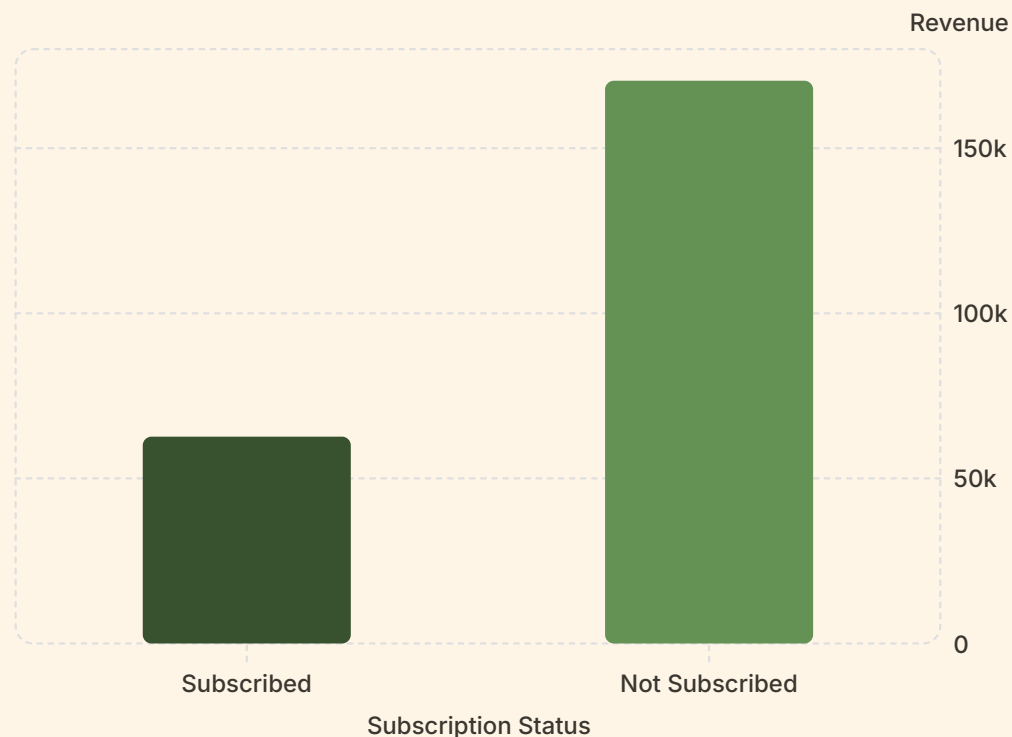
Hat

Avg Rating: **3.80**

Skirt

Avg Rating: **3.78**

Subscriber vs. Non-Subscriber Behavior



Key Insights

- **2,847 non-subscribers** generated \$170,436 in revenue
- **1,053 subscribers** generated \$62,645 in revenue
- Average spend is nearly identical (~\$59.50 vs. ~\$59.87)
- Non-subscribers represent the larger customer base

📄 Subscribers are **repeat buyers** — 958 subscribed buyers vs. 2,518 non-subscribed.

CUSTOMER SEGMENTATION

Customer Segments: New, Returning & Loyal

3116

Loyal Customers

The dominant segment —
highest repeat purchase
frequency

701

Returning Customers

Making additional purchases
but not yet fully loyal

83

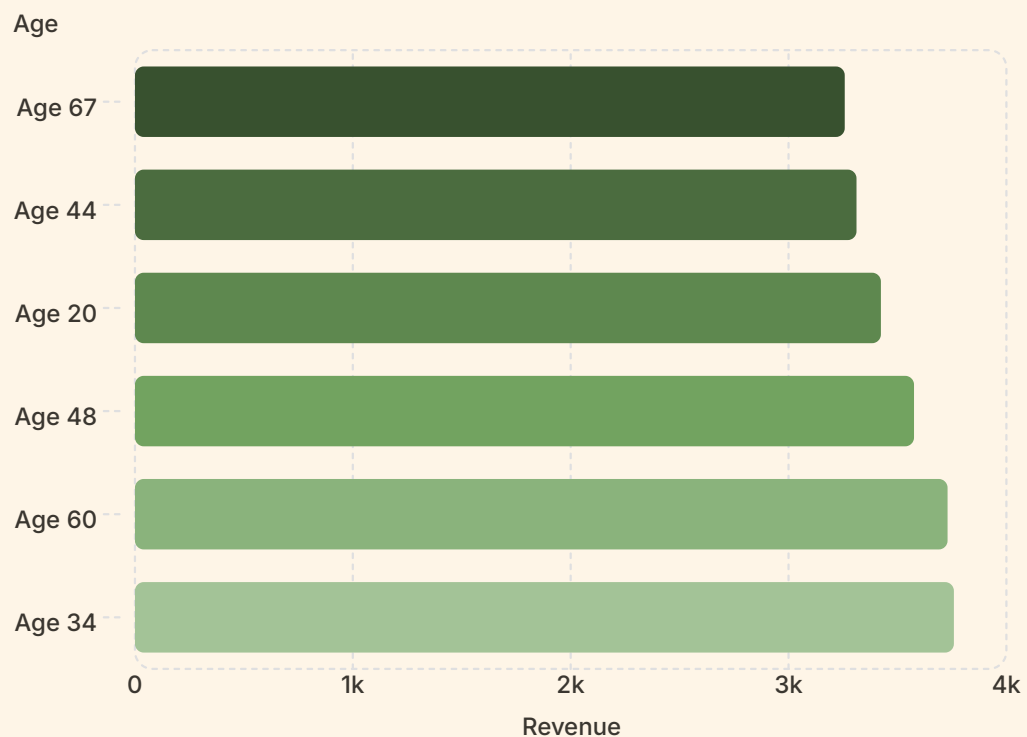
New Customers

First-time buyers — key
growth opportunity

Loyal customers represent **80%** of the customer base, indicating strong retention. New customers (83) represent the smallest segment and the primary growth opportunity.



Revenue Contribution by Age



Key Finding

Revenue is relatively evenly distributed across age groups, with **Age 34** contributing the highest (\$3,759) and **Age 67** the lowest (\$3,258).

Young adults and middle-aged customers lead in both revenue and customer count, making them the most valuable demographic segments.

VISUALIZATION

Power BI Dashboard Highlights

Subscription Status

92.7% non-subscribers vs. 7.3% subscribers

Revenue by Category

Clothing leads (~\$0.08M), followed by Accessories, Footwear & Outerwear

Revenue by Age Group

Young adults generate the highest revenue (~\$55K), followed by middle-aged, adult, and senior

Avg Purchase Amount

\$59.76 average across 3,900 transactions with 3.75 avg review rating



Power BI Dashboard Highlights

This interactive Power BI dashboard provides a comprehensive overview of customer shopping behavior. It consolidates key insights from the analysis, allowing stakeholders to easily track trends, identify top-performing products, understand customer segmentation, and monitor the impact of discounts and subscription models. The dashboard empowers data-driven decision-making with dynamic filters and drill-down capabilities.

